Pre-registration: Where the Universe Remembers — §6 Protocol

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This file constitutes the pre-registration for the observational test specified in §6 of *Where the Universe Remembers: A Geometric-Memory Reading of Dark Matter* (Zenodo, 2026). The specification below is fixed as of this release and will not be adjusted after contact with data. It accompanies the July 2026 prediction-audit record and prior-art survey record in this repository.

The claim

If the collisionless channel keeps event-specific records, then clusters that experienced a major merger and have since *genuinely settled* should retain residual structure in their mass distribution after their gas — the weave — has relaxed and forgotten. Standard cosmology's own simulations calibrate the opposing null with unusual sharpness: in ΛCDM, once a cluster is genuinely relaxed, the offset between its total-mass centroid and its light is essentially zero, with the residual floor set by reconstruction systematics rather than by physics. A near-zero null is the sharpest kind: the test can lose cleanly.

Primary statistic

The projected offset between the lensing total-mass centroid and the stellar light (BCG) centroid, measured in combined strong+weak lensing reconstructions, with the centroid definition fixed to track the reconstruction's potential minimum and the systematics floor calibrated per-system in mock reconstructions.

Secondary consistency statistic: very low-order mass-map asymmetry against settlement-matched controls.

Multipole/power-spectrum residuals are designated exploratory only: their time-domain null is not yet calibrated in the literature and a persistence claim cannot responsibly be built on them.

Settlement criterion — a three-part conjunction, no member optional

1. Low X-ray centroid shift — the morphology measure most robust to resolution and feedback systematics.
2. An established cool core, by central entropy or central cooling time — a thermodynamic criterion, slow to restore and hard to fake in projection.
3. BCG–X-ray peak consistency, applied last, to remove sloshing and miscentred systems.

Single-indicator definitions of "relaxed" are explicitly rejected; individual morphological parameters overlap between relaxed and disturbed populations (Campitiello et al. 2022, arXiv:2205.11326).

Test population

Systems satisfying the conjunction *and* carrying independent fossil evidence of a past major merger — radio relics being the natural timestamp, as they outlive the gas disturbance that produced them.

Control population: settlement-matched systems with no fossil merger evidence.

The designed-against mimic, named

ΛCDM itself produces persistent centroid displacement in unequal (3:1, 10:1) mergers, where surviving remnant substructure keeps the mass centroid moving past formal relaxation (Poole et al. 2006, arXiv:astro-ph/0608560). This is part of the null, not the signal, and it is the confounder most likely to counterfeit a detection. The settlement conjunction — particularly the cool-core and peak-consistency members — is constructed to exclude the regime in which this mimic operates, and any candidate excess must additionally be shown not to correlate with detectable substructure in the lensing map itself.

Null anchors, named

- **MACS J1423.8+2404** ($z = 0.545$): the most relaxed massive cool-core cluster known at $z > 0.5$; combined strong+weak lensing shows a strictly unimodal mass distribution with total-mass, stellar-light, and X-ray centroids in excellent alignment (Patel et al. 2024, arXiv:2405.04577); an active JWST deep-field target (CANUCS: Sarrouh et al. 2025, arXiv:2506.21685).
- **Abell 383** ($z = 0.1871$): relaxed, cool-cored, measured lensing-to-X-ray offset 1.0″ = 3.13 kpc, statistically indistinguishable from zero (offset tabulated in Shan et al. 2010, arXiv:1004.1475; relaxed status and combined modelling in Newman et al. 2011, arXiv:1101.3553).
- **Abell S1063** is excluded from any baseline role: residual X-ray brightness structure and a giant radio halo identify an ongoing merger.

The window, honestly sized

Current weak-lensing centroid uncertainties in even well-studied systems are of order tens of kiloparsecs, against a sought residual plausibly of a few to a few tens of kiloparsecs. Most existing archives therefore cannot run this test; only JWST-era deep combined reconstructions approach the regime. The null anchors above are already mapped at qualifying depth, so the test requires no instrument that does not exist — only the pointing of existing depth at the fossil-merger population. This pre-registration claims no result. It fixes the protocol — statistic, null, conjunction, mimic, controls, anchors — before the data quality arrives, so that when qualifying systems are measured, neither this framework nor its critics can move the goalposts.

Kill condition, both directions

If settled, fossil-merger clusters at qualifying data quality show mass–light offsets consistent with the systematics-calibrated null, the address channel does not keep event records at observable amplitude, and the reading fails at cluster scale. The author commits to reporting that outcome under the paper's title.

If a statistically secure excess appears, confined to the fossil-merger population and uncorrelated with detectable substructure, standard cosmology has an anomaly with a specified shape — and this reading predicted where it would be found.

Withdrawn predictions (documented negatives, per programme policy — see §5 of the paper): the cosmological "memory quantum" (no discretisation exists to inherit it from); the slow-accumulation prediction (untestable — "never-collided" is not an observable category); the event-erasure mechanism (never derivable from the framework; replaced by the weave-only reading, which requires no erasure).